

HOMEWORK 11
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

SECTION 4.2 - RESIDUE THEOREM

1. (a) Evaluate the contour integral $\int_{\gamma} \frac{e^{-z^2}}{z^2} dz$, where γ is the square with corners $-1-i, 1-i, 1+i, -1+i$.
(b) Evaluate the contour integral $\int_{\gamma} \frac{e^{-z^2}}{z^2} dz$, where γ is the ellipse $\gamma(t) = a \cos t + ib \sin t$ for $t \in [0, 2\pi]$ and $a, b > 0$.
2. Let f be analytic on $\mathbb{C}$ except for poles at 1 and -1 . Assume that $\text{Res}(f; 1) = -\text{Res}(f; -1)$. Let $A = \{z : z \notin [-1, 1]\}$. Show that there is an analytic function h on A such that $h'(z) = f(z)$.
3. (a) Find the residue of $\frac{(z-1)^3}{z(z+2)^3}$ at $z = \infty$.
(b) Give two methods of evaluating

$$\int_{\gamma} \frac{(z-1)^3}{z(z+2)^3} dz,$$

where γ is the circle with center zero and radius 3.

SECTION 4.3 - EVALUATION OF DEFINITE INTEGRALS

4. Evaluate $\int_0^{\pi} \frac{d\theta}{(a+b \cos \theta)^2}$ for $0 < b < a$.
5. Evaluate $\int_0^{\infty} \frac{x^{a-1}}{1+x^3} dx$ for $0 < a < 3$.
6. Show that $\int_0^{\pi} \sin^{2n} \theta d\theta = \frac{\pi(2n)!}{(2^n n!)^2}$.